

ATIN SHARMA

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PROFILE

Full-Stack Product Engineer with founding-team experience building AI systems for Indian SMBs.

- Shipped VyaparGPT — WhatsApp-native LLM assistant (40 SMB pilot)
- Reduced API latency by 40% via PostgreSQL optimization (N+1 fixes, caching)
- Built production RAG pipelines (pgvector, sub-800ms retrieval)

PROFESSIONAL EXPERIENCE

OpenBiz Software India Pvt Ltd

Jun 2025 – May 2026

Founding Engineer

Remote / Bangalore

- VyaparGPT: built full WhatsApp message lifecycle (webhook ingestion → HMAC-SHA256 validation → session context → LLM dispatch → response delivery), serving 40+ SMBs in closed pilot, scaling to 1,000+ active platform users. Multi-turn conversational context via rolling `conversation_history`.
- LLM provider-fallback: OpenAI → Google Gemini on HTTP 5xx, 429 rate-limit, or configurable timeout, with response-shape normalisation — zero failure surface exposed to users.
- Gemini Vision API document verification module: extracting structured fields (GST number, business name, invoice amount, dates) from WhatsApp photographs of GST certificates and vendor agreements, eliminating a manual data-entry step in SMB onboarding.
- N+1 fix: rewrote ORM calls to JOIN-based queries, introduced composite indexes on high-frequency WHERE/JOIN columns, and added TTL-based response caching — achieving verified **40% API latency reduction** at 1,000+ user scale.
- Razorpay: HMAC webhook validation, payment-state reconciliation, out-of-order event handling via re-entrant idempotency check — achieving **100% payment-state consistency**.
- CI/CD: GitHub Actions + Vercel, automated testing on every PR, staging deploys on merge, production gated behind explicit approval — **30% reduction in deployment cycle time** and zero broken-build incidents over four months.
- Website Builder product: shipped within 6-week window at zero headcount increase by auditing and reusing ~60% of shared infrastructure (auth, Supabase layer, Razorpay billing, CI/CD).

Vellore Institute of Technology

Jan 2025 – Nov 2025

Research Contributor

Vellore (On-campus)

- Contributed to research on **Blockchain-Based LLM Model Using Fully Homomorphic Encryption (FHE) for Academic Records** — investigating how FHE enables computation on encrypted student data without decryption, with blockchain providing an immutable audit trail for credential issuance and verification.
- Evaluated architectural trade-offs between on-chain data storage and off-chain encrypted record pointers, assessing gas cost constraints versus data privacy guarantees in an academic credential provenance system.
- Explored how model inference could operate on privacy-preserving representations of sensitive academic data using FHE-compatible cryptographic primitives, without exposing plaintext records.

PROJECTS

Nimbus — AI RAG Workspace | Next.js 14, TypeScript, PostgreSQL, pgvector, Prisma, NextAuth.js, Groq API, OpenAI | In Development 2026

- Production-grade RAG pipeline: async queue-based ingestion (status states: pending → processing → indexed → error), overlap chunking to prevent semantic loss at chunk boundaries, embedding generation, and storage in a pgvector extension column within PostgreSQL.
- Cosine similarity search via pgvector (`embedding <=> query_vector`) co-locates vector data with relational metadata (`user_id, document_id`) for hybrid SQL + semantic filtering — sub-800ms retrieval target.
- Provider-agnostic LLM abstraction layer: single config flag (`provider: "groq" | "openai" | "anthropic"`) routes generation calls — switching inference providers requires zero code changes.
- Streaming responses via Groq API delivered as SSE, providing real-time token-by-token document Q&A. Ingestion pipeline includes 429 retry logic (exponential backoff) and per-document status tracking.

DocGPT — RAG Document Assistant | React 19, Python (Django), ChromaDB, OpenAI, Docker

- Full Django RAG pipeline: PDF extraction via PyPDF2 with overlap chunking (~500–1000 tokens, ~50–100 overlap), **local embedding** via all-MiniLM-L6-v2 (384-dimensional, offline — zero API cost), ChromaDB cosine retrieval, GPT-4o-mini generation.
- Containerized Django backend + ChromaDB as Docker services; prompt structure: system context + k-nearest retrieved chunks + user query for grounded generation.

Autonomous Web Agent | github.com/atinsharma24/auto-agent | Python, Groq API, BrowserOS (MCP), Claude 3.5 Sonnet

- Dual-agent architecture: Executor (Claude 3.5 Sonnet) handles complex multi-step DOM + form reasoning; Watcher (Groq fast inference) validates page state between every transition — separating expensive reasoning from high-frequency cheap polling.
- MCP (Model Context Protocol) via BrowserOS gives the Executor structured tool-level access to browser DOM (inspect labels, types, values), form-fill, and navigation — intent-driven traversal with no brittle CSS selectors.

- Persistent local queue manager (status: pending → in_progress → applied / failed / escalated) with URL deduplication; WAF/CAPTCHA detection via DOM fingerprint – Watcher escalates to human, Executor resumes from last checkpoint.

Automated Content Intelligence Pipeline | *React 19, TypeScript, Laravel, Node.js, OpenAI, Gemini, Docker, GitHub Actions, Puppeteer*

- Puppeteer headless Chromium (`waitUntil: 'networkidle2'`) for SPA scraping; dual-model AI rewriting (Gemini primary → OpenAI fallback per article); React 19 split-screen Diff View with per-article state machine (pending_review → approved → published).
- Dockerized microservices (Scraper, AI Rewriter, Laravel API, React Frontend) with unless-stopped restart policy; GitHub Actions cron schedule drives fully automated ingestion and review cycles.

SKILLS

Languages: JavaScript/TypeScript, Python, SQL, Java

Frontend: React (v18/19), Next.js, React Native, Redux Toolkit, TailwindCSS

Backend/DB: Node.js, Express, NestJS, Django, PostgreSQL, MongoDB, pgvector, ChromaDB

AI/LLM: OpenAI, Google Gemini, Groq API, RAG Pipelines, LangChain.js, Sentence Transformers, Prompt Engineering

DevOps: Docker, GitHub Actions, Vercel, AWS (S3, DynamoDB)

Tools: Git, Postman, Socket.IO, Razorpay SDK, VideoSDK, Puppeteer

Patterns: WebSocket (Socket.IO), REST, Webhook Validation (HMAC-SHA256), N+1 Query Optimization, Vector Similarity Search (cosine), Provider-Agnostic LLM Layers, Idempotent Event Processing

EDUCATION

Vellore Institute of Technology

B.Tech in Computer Science and Engineering

- Focus on Full Stack Development and AI Systems.

Expected Graduation: May 2026

Vellore

PROFILE SNAPSHOT

Availability: Immediate Joiner – 0-day notice period

Work Mode: Remote-first; open to Bangalore, Hyderabad, Pune, Delhi NCR, Mumbai

Expected CTC: 15–18 LPA (INR) – negotiable for high-ownership roles with equity

Work Authorization: Indian Citizen – no sponsorship required

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